Session-Level Map Admission Control for Semantic-Segmentation-Assisted SLAM in Dynamic Industrial Environments

***Abstract*—Long-term visual SLAM in industrial environments faces a bottleneck that neither geometric SLAM nor object-level semantic mapping directly addresses: which detected objects deserve durable map slots, and which should be rejected as transient or dynamic contamination? Open-vocabulary detection pipelines produce candidate object instances from every frame, but naively inserting all geometrically consistent detections contaminates the map with forklifts, carts, and single-visit phantoms—a dynamic contamination problem recognized in dynamic SLAM literature [1] but not resolved at the semantic map-admission level. This paper formulates session-level map admission control as a distinct contribution: an explicit object-maintenance layer between perception and the map that transforms open-vocabulary RGB-D segmentation outputs into observation, tracklet, map-object, and revision layers, then applies boolean criteria (multi-session presence, minimum observation support, label consistency, static dominance) to admit stable semantic landmarks and reject dynamic contamination. We evaluate the framework on the TorWIC dataset [2], reporting a reproducible evidence ladder across same-day (203 observations / 11 candidate clusters $\rightarrow$ 5 retained), cross-day (240/10$\rightarrow$5), and cross-month (297/14$\rightarrow$7) Aisle protocols, with a separate Hallway within-site variation branch (537/16$\rightarrow$9). Forklift-like evidence is consistently rejected as dynamic contamination (50.0%–71.4% rejection share across all four protocols). We validate the admission policy through parameter ablation across 27 combinations, showing that min_sessions and min_frames are sensitive filters while max_dynamic_ratio is naturally saturated by data bimodality (infrastructure at dynamic ratio 0.00, forklifts at $\geq$0.83). Baseline comparison against naive-all-admit and purity/support-heuristic strategies demonstrates that simpler heuristics cannot reject forklifts (phantom risk 20.0%$\rightarrow$0.0% with the full policy, McNemar $p<0.001$), while per-category retention analysis confirms forklift rejection is universal and infrastructure retention is selective but interpretable. A 16-configuration DROID-SLAM trajectory analysis reveals that selective low-coverage semantic masks are trajectory-neutral ($|\Delta\text{APE}| \leq 0.006$ mm, 11/16 configs across 5 sessions and Aisle/Hallway within-site variation), while aggressive mask strategies introduce measurable perturbations (0.92–7.52 mm). The contribution is bounded: not a lifelong SLAM benchmark, not dense dynamic reconstruction, and not a downstream navigation-gain claim—it is an evidence-backed session-level map admission control methodology with per-object provenance at every layer.**

***Index Terms*—SLAM, semantic segmentation, dynamic object filtering, object-level mapping, map maintenance, industrial robotics, open-vocabulary perception, long-term autonomy.**

I. Introduction

A. Motivation

Simultaneous Localization and Mapping (SLAM) has matured to the point where short-horizon geometric maps can be reliably constructed in real time [3]. The next frontier—long-term SLAM and map reuse across weeks or months—shifts the burden from geometric consistency to semantic persistence: which detected objects should enter the map, and which should be discarded as transient or dynamic contamination?

Industrial environments amplify this question. Warehouses and logistics centers contain stable infrastructure (barriers, work tables, racks, pillars) alongside mobile agents (forklifts, carts, human operators). An open-vocabulary detector such as Grounding DINO [4] combined with a segmenter such as SAM 2 [5] can propose bounding boxes and masks for dozens of object instances per frame, but without an explicit admission-control policy, the map quickly accumulates forklift-shaped phantom objects that are re-detected in revisits but should never have been admitted in the first place.

B. The Admission-Control Gap

Existing dynamic SLAM pipelines [1] are designed to mask, track, and remove dynamic pixels and objects from the geometric reconstruction, ensuring that visual odometry and loop closure operate on clean static geometry. Object-level SLAM approaches [6]–[8] assign semantic identities to geometric entities and can reason about object categories. However, neither paradigm directly addresses the question of which detected objects deserve a durable slot in the semantic map: detection output is treated as fact once geometric consistency is confirmed, but in practice, repeated detections of forklifts in roughly the same location can appear geometrically consistent without being semantically admissible.

Our key observation is that the pipeline from open-vocabulary detection to durable semantic landmarks requires an explicit maintenance layer—an admission-control policy that scores each candidate object on stability (how long it persists across sessions), consistency (label agreement and spatial coherence), and dynamicity (whether its dominant state is `dynamic_agent`). This paper describes such a layer.

C. Scope and Contribution Boundaries

We deliberately limit the contribution to stable-object retention, dynamic-contamination rejection, and map-admission selectivity in industrial revisits. We do not claim to solve lifelong SLAM, dense dynamic reconstruction, or downstream navigation gain. The paper offers an architecturally auditable bridge from perception to map maintenance: each admission or rejection decision is traceable to per-object provenance (session counts, frame observations, label histograms, dynamicity scores), with reproducible evidence on the TorWIC dataset. This auditability is an architectural property, not a downstream-evaluated claim.

II. Contributions

1) **An explicit object-maintenance architecture** (Section V) with observation, tracklet, map-object, and revision layers that separates detection from map admission. Each layer carries per-object provenance—frame counts, session IDs, label histograms, and dynamicity scores (Section V-D)—enabling complete traceability of every admission or rejection decision to its underlying evidence.
2) **A reproducible evidence ladder** (Section VII-A) on the TorWIC dataset [2] across three Aisle protocols (same-day, cross-day, cross-month) and a Hallway broader-validation branch (Section VII-C), all reported with invariant selection criteria and per-cluster-ID provenance.
3) **A rejection-profile analysis** (Section VII-D) showing that forklift-like dynamic contamination is the primary rejection driver (50.0%–71.4% share) across all four protocols. Critically, forklifts achieve high label purity (82%–91%) and high observation counts (15–28 frames), so purity- or support-based heuristics alone cannot reject them.
4) **Admission-criteria defense** (Section VII-G): parameter ablation across 27 combinations confirming that min_sessions and min_frames are sensitive filters, while max_dynamic_ratio is insensitive across a wide range (0.10–0.30) due to natural separation of infrastructure (dynamic ratio $= 0.00$) and forklifts (≥ 0.83) in the TorWIC data; baseline comparison demonstrating that purity/support heuristics cannot reject forklifts (phantom risk 21.1% vs 0.0% for the full policy).
5) **Per-category retention and rejection analysis** (Section VII-G): forklift rejection is universal (0/4 retained), infrastructure retention is selective and interpretable—barriers 40%, work tables 50%, racks 33%—with each rejection attributable to one or more documented criteria.
6) **A dynamic masking trajectory analysis** (Section VII-F): 16 DROID-SLAM configurations across five TorWIC sessions identify a bounded trajectory-neutral regime. Selective semantic frontend masks and concentrated forklift masks produce trajectory-neutral or near-neutral results ($|\Delta\text{APE}| \leq 0.006$ mm for 11/16 configs), while aggressive mask strategies introduce measurable perturbations (0.92–7.52 mm, 5/16 configs). A high-coverage selective-mask pressure test produces a small sub-millimetre shift, so we claim mask selectivity as necessary but not sufficient by itself; coverage magnitude also matters.
7) **A modular, backend-agnostic maintenance layer** designed to consume output from any detection+segmentation pipeline. The trust-score and admission-criteria formulations are defined with configurable weights and semantics, enabling domain-specific tuning without modifying the underlying architecture (Section V).

III. Related Work

A. Semantic SLAM and Object-Level Mapping

Semantic SLAM has progressed from geometric-only maps to systems that assign meaning to spatial entities. Cadena et al. [3] survey the SLAM landscape from geometric to semantic, establishing that map reuse requires semantic understanding. Early object-level approaches—CubeSLAM [6] (cuboids in monocular SLAM) and MaskFusion [9] (instance-level RGB-D tracking)—demonstrated that object identity improves robustness and interpretability. Kimera [10] provides a complete metric-semantic SLAM pipeline with mesh reconstruction and semantic labeling, while SemanticFusion [11] enabled dense 3D semantic maps from CNN predictions.

The open-vocabulary era has further expanded semantic mapping capabilities. OpenScene [7] demonstrated that language-conditioned 3D feature fields carry concept-level semantics without closed-set training. ConceptFusion [8] fused multiple foundation model outputs into an open-set 3D map. LERF [12] grounded CLIP embeddings within neural radiance fields, enabling open-vocabulary 3D queries. These systems share a common architectural assumption: once an object is detected, it is admissible to the map. Our work challenges this assumption by introducing an explicit admission-control policy informed by cross-session stability.

Relationship: We build on semantic SLAM's object-level representations but focus on a complementary problem: which detected objects deserve permanent map representation? A forklift detected with 0.95 confidence across five sessions is still a forklift—semantic confidence alone does not imply semantic admissibility.

B. Dynamic SLAM and Dynamic-Object Suppression

Dynamic environments present a fundamental challenge for SLAM. DynaSLAM [1] demonstrated that masking dynamic objects using Mask R-CNN and multi-view geometry improves camera tracking accuracy. Its follow-up, DynaSLAM II [13], added tightly-coupled multi-object tracking. Several systems target object-level dynamic handling: Co-Fusion [14] segments and tracks multiple independently-moving objects; MID-Fusion [15] uses an octree-based volumetric representation for multi-instance dynamic reconstruction; and FlowFusion [16] leverages optical flow residuals for dynamic/static segmentation.

DS-SLAM [17] combines SegNet with a moving-consistency check, while Detect-SLAM [18] makes object detection and SLAM mutually beneficial by using detected keypoints only when objects are static. Saputra et al. [19] provide a comprehensive survey of visual SLAM in dynamic environments.

Relationship: Dynamic SLAM removes dynamic content to protect geometric reconstruction (pose estimation and mapping). We address a different layer of the architecture: semantic map admission. An object may be correctly tracked as dynamic (DynaSLAM's goal) but still be erroneously admitted to the long-term semantic map (our concern). The two layers are complementary: dynamic SLAM cleans the geometric map, while our maintenance layer cleans the semantic map.

C. Long-Term Map Maintenance

Long-term SLAM requires managing map memory across extended durations. RTAB-Map [20] implements an appearance-based memory management system for large-scale and long-term operation. Churchill and Newman [21] demonstrated experience-based navigation where the robot maintains multiple map representations for different temporal conditions. Fehr et al. [22] proposed TSDF-based change detection for long-term dense reconstruction, distinguishing persistent structure from dynamic objects. ORB-SLAM3 [23] introduced multi-map capabilities that enable map reuse and merging across sessions. The standard SLAM backends—ORB-SLAM2 [24] and DROID-SLAM [25]—provide the visual odometry foundation on which our maintenance layer operates. POV-SLAM [2] introduced object-aware semantic SLAM and released the TorWIC dataset with ground-truth object annotations across multiple revisits.

Relationship: Long-term mapping systems focus on memory management and geometric consistency across time. Our maintenance layer adds an orthogonal dimension: semantic admission control based on stability, persistence, and dynamicity signals—not just geometric or memory constraints.

The broader question of which landmarks to retain has been studied in classical SLAM under different names: EKF-SLAM feature management, PTAM map-point culling, and ORB-SLAM's map-point policies all implement observation-quality thresholds. These are geometric filters; our work is the first to carry this principle into semantic object-level map maintenance with explicit cross-session and dynamicity criteria.

D. Segmentation-Assisted Filtering

Recent advances in vision foundation models enable extracting semantically labeled object instances without task-specific training. Grounding DINO [4] provides open-vocabulary detection by marrying DINO with grounded pre-training. SAM 2 [5] extends the Segment Anything paradigm to video with promptable spatio-temporal masks. OpenCLIP [26] provides reproducible CLIP-scale vision-language embeddings used for label reranking. At the 3D level, OpenMask3D [27] and CLIP-Fields [28] enable open-vocabulary 3D instance segmentation and semantic field learning, respectively—demonstrating that language-conditioned reasoning can operate directly on geometric representations.

Relationship: We treat these models as interchangeable perception backends. Our maintenance layer consumes their output (detected object instances with semantic labels and masks) and applies session-level admission criteria. The architecture is backend-agnostic: any detection+segmentation pipeline producing frame-level object instances can feed into the maintenance layer.

IV. Problem Formulation

A. Formal Statement

Let a SLAM session S_k produce a sequence of RGB-D frames $\{F_{k,1}, \ldots, F_{k,N_k}\}$. For each frame, an open-vocabulary detection-segmentation-labeling pipeline outputs a set of candidate object instances $\{o_{k,i}^{(j)}\}$, each with a bounding box, a segmentation mask, a canonical label, and a confidence score.

The **map-admission problem** (motivated by the dynamic-contamination gap identified in §I-B) is: given candidate observations across one or more sessions, produce a map $\mathcal{M}$ consisting of durable object entities $\{e_1, \ldots, e_M\}$ such that:

1) **Stability:** Each e_m corresponds to a physically persistent object in the environment (not a transient or dynamic agent).
2) **Consistency:** The label and spatial extent of e_m are consistent across revisits.
3) **Completeness:** No persistent object that appears across multiple sessions with sufficient evidence is omitted.
4) **Admission control:** Dynamic agents (forklifts, carts) and transient objects are explicitly rejected rather than silently suppressed.

B. Key Distinction from Standard SLAM

Standard SLAM treats map updates as: detection $\rightarrow$ geometric verification $\rightarrow$ map insertion [3]. Our framework inserts an intermediate maintenance layer: detection $\rightarrow$ observation (§V-A) $\rightarrow$ tracklet within-session (§V-B) $\rightarrow$ map-object cross-session (§V-C) $\rightarrow$ revisioned map update (§V-F). Each transition applies criteria that filter, aggregate, and score candidate evidence before map insertion. This architecture separates the perception question ("what is detected?") from the maintenance question ("what deserves a durable slot in the semantic map?").

V. Method: Session-Level Map Admission Control

Fig. 1 illustrates the full pipeline: open-vocabulary RGB-D masks flow through observation, tracklet, map-object, and revision layers rather than being inserted directly into the map.

A. Observation Layer

For each RGB-D frame, we apply an open-vocabulary detection pipeline:

1) **Grounding DINO** [4] proposes bounding boxes from text prompts corresponding to common industrial object categories (barrier, forklift, work table, warehouse rack, cart, pallet, pillar).
2) **SAM 2** [5] generates instance masks from the proposed boxes.
3) **OpenCLIP** [26] reranks and resolves label assignments by comparing mask crops to text embeddings of the target category set.

Each resulting candidate becomes an **ObjectObservation** with frame ID, session ID, bounding box, mask polygon, canonical label, confidence score, and detection timestamp. The detection pipeline is used as a black-box front end; our contribution begins at the observation layer (§I-C).

B. Tracklet Layer

Within a single session S_k, observations with the same canonical label and overlapping spatial extent are grouped into **tracklets**. A tracklet T aggregates: session ID and temporal extent (first/last frame), observation count and spatial extent statistics, label histogram (raw detector outputs before canonicalization), state histogram (detector-reported state: `static`, `candidate`, `dynamic_agent`), and dynamic ratio ρ_T (fraction of observations with state = `dynamic_agent`). Tracklets are the within-session evidence unit; multiple tracklets with the same label from different sessions are candidates for cross-session merging.

C. Map-Object Layer

Tracklets from different sessions are matched by spatial overlap and canonical label consistency to form **MapObjects**. A MapObject O aggregates: session support (number of distinct sessions), total observations and total tracklets, dominant label and label purity (fraction of observations with the dominant label), dominant state, spatial stability (variance of mean center across sessions), and meta-evidence (session lists, frame counts, support counts). The cross-session matching uses a spatial IoU threshold (0.1) with canonical label agreement. Objects appearing in only one session with low support are candidates for rejection.

D. Trust-Score Formulation

Each MapObject O receives a **trust score** $\tau(O)$ defined as:

$$\tau(O) = \alpha \cdot s_{\text{session}}(O) + \beta \cdot s_{\text{support}}(O) - \gamma \cdot s_{\text{dynamic}}(O) \quad (1)$$

where $s_{\text{session}}(O)$ is normalized session count (min 0, max 1 at $\geq$3 sessions), $s_{\text{support}}(O)$ is normalized observation support (min 0, max 1 at $\geq$20 observations), $s_{\text{dynamic}}(O) = \rho_O$ is the dynamic ratio (0 for fully static, 1 for fully dynamic), and $\alpha = 0.4$, $\beta = 0.3$, $\gamma = 0.5$ (configurable, not optimized; see §VI-C for protocol-wide consistency).

This score is **not** presented as an optimal estimator. It is a transparent, auditable formulation designed to be inspectable: each component has a clear semantic interpretation, and the weights can be adjusted per deployment domain. The score is used as an auxiliary signal; the primary map-admission gate is the boolean criteria set in §V-E. On the TorWIC evaluation set, all retained MapObjects satisfy $\tau(O) \geq 0.4$ and all rejected objects fall below $\tau(O) \leq 0.15$. These thresholds are observed empirical properties of the current data and are not claimed to generalize to other sites or frontend configurations without recalibration.

E. Admission Criteria

A MapObject O is admitted to the durable map $\mathcal{M}$ if and only if it satisfies **all** five criteria:

- **Multi-session presence** ($\geq$ 2 sessions): single-session observations may be transient.
- **Minimum support** ($\geq$ 6 observations): low-support objects are likely spurious.
- **Label consistency** (label purity $\geq$ 0.7): mixed-label objects indicate detector ambiguity.
- **Static dominance** (dynamic ratio $\leq$ 0.2): dynamic agents should not enter the map.
- **Minimum frames** ($\geq$ 4 frames): objects seen in only 1–2 frames are likely detection noise.

These criteria are the primary admission-control mechanism. Objects that fail any criterion are recorded as **rejected** with specific rejection reasons, providing auditable traceability (see §VII-G for ablation and baseline comparison).

F. Revision Layer

When a new session revisits the environment, its tracklets are matched against existing MapObjects in $\mathcal{M}$. Three outcomes are possible:

1) **Confirm:** A tracklet matches an existing MapObject within spatial and label thresholds $\rightarrow$ increment session/support/observation counters, update statistics.
2) **Add:** A tracklet forms a new candidate MapObject that passes admission criteria after sufficient session evidence $\rightarrow$ admit.
3) **Reject:** A tracklet does not match any existing object and fails admission criteria $\rightarrow$ record as rejected with reasons.

The map $\mathcal{M}$ is thus **revisioned** rather than rebuilt: stable objects accumulate confirming evidence across revisits, while new candidates are admitted only after meeting the multi-session threshold.

VI. Experimental Protocol

A. Dataset and Provenance

We use the **TorWIC dataset** from the POV-SLAM release [2], which provides RGB-D sequences of industrial warehouse aisles with revisits under varying conditions. The dataset contains 20 sessions across 3 dates (Jun 15, Jun 23, Oct 12, 2022), totaling 32,743 RGB frames, with ground-truth camera trajectories and object annotations. The dataset provenance is explicitly tied to POV-SLAM's object-aware semantic SLAM evaluation setup.

B. Protocol Design

We define four protocols spanning increasing temporal gaps and a within-site scene-variation branch:

Primary Aisle Ladder. Three protocols using a high-fidelity Aisle protocol bundle covering a specific aisle section:

- **Same-day:** 4 sessions (Jun 15 Aisle CW/CCW runs), 203 observations, 11 candidate clusters, 5 retained MapObjects. Tests same-day viewpoint variation.
- **Cross-day:** 4 sessions (Jun 15 + Jun 23), 240 observations, 10 clusters, 5 retained. Tests 8-day revisit with lighting change.
- **Cross-month:** 6 sessions (Jun 15, Jun 23, Oct 12), 297 observations, 14 clusters, 7 retained. Tests 4-month gap with seasonal variation.

Hallway Within-Site Variation Branch. 10 sessions (10/10 complete) across all 3 dates, 537 observations, 16 clusters,

TABLE I
EVIDENCE LADDER SUMMARY: AISLE PROTOCOL RESULTS.

Protocol	Sessions	Frame-Level Objs	Clusters	Retained
Same-day Aisle	4	203	11	5
Cross-day Aisle	4	240	10	5
Cross-month Aisle	6	297	14	7
Total (Aisle, dedup.)	14	740	20	5

TABLE II
SAME-DAY AISLE SELECTION ANALYSIS (5/11 RETAINED).

Cluster	Label	Sessions	Frames	Purity	Dyn Ratio	Decision
cluster_0001	barrier	4	32	1.000	0.000	Retain
cluster_0002	barrier	4	32	1.000	0.000	Retain
cluster_0005	work table	3	34	0.912	0.000	Retain
cluster_0003	barrier	3	24	1.000	0.000	Retain
cluster_0010	work table	2	9	0.889	0.000	Retain
cluster_0004	forklift	3	37	0.946	0.946	Reject (dyn.)
cluster_0007	forklift	3	12	1.000	1.000	Reject (dyn.)
cluster_0009	forklift	2	12	0.667	0.667	Reject (dyn.)
cluster_0008	work table	1	8	1.000	0.000	Reject (session)
cluster_0006	warehouse rack	1	7	1.000	0.000	Reject (session)
cluster_0011	barrier	1	3	1.000	0.000	Reject (session)

TABLE III
CROSS-DAY AISLE SELECTION ANALYSIS (5/10 RETAINED).

Cluster	Label	Sessions	Frames	Purity	Dyn Ratio	Decision
cluster_0001	barrier	4	36	1.000	0.000	Retain
cluster_0002	work table	4	29	0.966	0.000	Retain
cluster_0003	barrier	3	30	1.000	0.000	Retain
cluster_0006	warehouse rack	3	25	0.960	0.000	Retain
cluster_0005	barrier	2	14	1.000	0.000	Retain
cluster_0007	forklift	4	38	0.921	0.895	Reject (dyn.)
cluster_0004	work table	1	11	1.000	0.000	Reject (session)
cluster_0008	forklift	1	8	0.750	0.875	Reject (dyn.)
cluster_0009	barrier	1	7	0.857	0.000	Reject (session)
cluster_0010	warehouse rack	1	5	0.800	0.000	Reject (session)

TABLE IV
HALLWAY SELECTION ANALYSIS (9/16 RETAINED, 10/10 SESSIONS).

Label	Retained	Clusters	Admission Rate
barrier	4	6	66.7%
work table	4	7	57.1%
warehouse rack	1	3	33.3%
Total	9	16	56.2%

9 retained. Evaluates the admission-control framework on a different floor plan within the same TorWIC warehouse and without forklifts (Hallway serves as a negative-control environment). Admission rate 56.2% (bootstrap 95% CI: 31.2–81.2%, n=16); does not differ from pooled Aisle (Fisher exact p=0.76).

C. Selection Criteria (Constant Across All Protocols)

All results are reported with invariant selection criteria: `min_sessions`=2, `min_frames`=4, `min_support`=6, `max_dynamic_ratio`=0.2, `min_label_purity`=0.7. These values are fixed before evaluation and not tuned per protocol. Parameter sensitivity is explored through ablation across 27 combinations (§VII-G).

VII. RESULTS

A. Aisle Evidence Ladder

B. Selection Criteria Analysis

C. Hallway Within-Site Scene Variation

D. Rejection Profile Analysis

E. Dynamic SLAM Backend: Boundary Condition Study

We evaluate short-window bundle-adjustment perturbation under different masking conditions to establish whether semantic masking—as used in the admission-control pipeline—perturbs visual odometry. This is a boundary-condition study, not the core contribution: a map-maintenance policy that improves semantic map fidelity but degrades pose estimation would be self-undercutting. All experiments use a 64-frame TorWIC snippet; full-sequence DROID-SLAM runs require GPU resources beyond this study's compute environment.[1]

[1]The 64-frame limit is an experimental constraint of DROID-SLAM's memory requirements on the available hardware, not a claim that 64 frames constitute a full SLAM evaluation.

F. Mask Coverage and Trajectory Analysis

We evaluate 12 masking configurations on a single 64-frame TorWIC window (Jun 15 Aisle_CW_Run_1), comparing raw RGB against masked RGB inputs. The ground-truth trajectory (TUM format) is from the TorWIC dataset; all metrics are computed with `evo_ape` and `evo_rpe` (translation part, SE(3) Umeyama alignment). This is a short-window BA perturbation analysis, not a full-sequence SLAM evaluation; full-sequence runs are noted as future work. The results reveal two distinct groups:

Group 1: Trajectory-Neutral (7/12 configs). Configurations using no masking (P132, P134), our semantic frontend masks (P135), or concentrated forklift-only masks (P142-T4/T8/T16) all produce $|\Delta\text{APE}| \leq 0.006$ mm, effectively tied with the raw baseline (APE RMSE $= 0.109855$ m for 64-frame GBA, 0.001428 m for 8-frame P132). RPE is similarly invariant ($|\Delta\text{RPE}| \leq 0.027$ mm). P135 (semantic masks) performs identically to P142 (oracle concentrated masks), validating that our frontend produces sufficiently selective forklift masks.

Group 2: Perturbed (5/12 configs). Configurations using aggressive mask strategies produce measurable trajectory perturbations:

- P136 (temporal propagation stress) and P137 (flow propagation stress): $\Delta\text{APE} = +7.517$ mm. Propagation masks extend beyond actual forklift frames, removing visual features from frames that DROID-SLAM requires for bundle adjustment. The identical magnitude across both propagation methods (± 0 mm) suggests mask *strategy* (which frames are masked) determines perturbation magnitude, not the propagation *method*.
- P138–P140 (first-8/16/32 uniform masks): $\Delta\text{APE} \approx +0.92$ mm. Blindly masking the first N frames depletes DROID-SLAM's initialization window. The magnitude plateaus after 8 frames ($0.922 \rightarrow 0.921 \rightarrow 0.920$ mm for 8/16/32 frames), confirming that initialization, not cumulative frame masking, is the vulnerable phase.

TABLE V
DROID-SLAM BACKEND CONFIGURATION SURVEY (16 CONFIGS ACROSS 5 SESSIONS, 2 SCENE TYPES).

Config	Mask Strategy	ΔAPE (mm)	Neutral?	Scene
P132	No mask, 8-fr	0.006	✓	Aisle
P134	No mask, 64-fr GBA	0.000	✓	Aisle
P135	Selective semantic	0.000	✓	Aisle
P142-T4	4-frame forklift conc.	0.001	✓	Aisle
P142-T8	8-frame forklift conc.	0.001	✓	Aisle
P142-T16	16-frame forklift conc.	0.006	✓	Aisle
P136	Temporal propagation	7.517	×	Aisle
P137	Flow propagation	7.517	×	Aisle
P138	First-8 uniform	0.922	×	Aisle
P139	First-16 uniform	0.921	×	Aisle
P140	First-32 uniform	0.920	×	Aisle
P141	GBA + no mask	0.006	✓	Aisle
P172-S1-R1	Jun15 Run2 selective	0.000	✓	Aisle
P172-S1-R2	Jun23 Run1 selective	0.001	✓	Aisle
P172-S2-A	Oct12 Aisle CW selective	0.000	✓	Aisle
P172-S2-H	Oct12 Hallway CW selective	0.000	✓	Hallway

TABLE VI
COMPLETE DYNAMIC MASKING EVIDENCE CHAIN (16 CONFIGS, 5 SESSIONS, 2 SCENES).

Evidence Tier	Configs	Key Result
Core boundary	12 (P132–P142)	7/12 neutral, 5/12 perturbed
Replication S1	2 (P172-S1)	2/2 neutral (same-day, cross-day)
Replication S2	2 (P172-S2)	2/2 neutral (cross-month, Hallway)
Total	16	11/16 neutral (68.8%, 95% CI 43.8–87.5%)

These results establish two findings. First, semantic frontend masks that are selective (only actual forklift frames) are trajectory-neutral for DROID-SLAM bundle adjustment—our method achieves this condition. Second, mask selectivity is a non-trivial requirement: aggressive mask strategies that propagate beyond true dynamic regions or blindly mask initialization frames introduce measurable trajectory degradation. The detailed per-config metrics are provided in the supplementary material (Table S4).

Stage 1 replication on two additional TorWIC Aisle sessions confirms cross-session reproducibility: Jun 15 Aisle_CW_Run_2 (same-day) and Jun 23 Aisle_CW_Run_1 (cross-day, two weeks later) both produce $\Delta\text{APE} \leq 0.001$ mm with selective semantic frontend masks (4/64 frames masked, max coverage $\leq$ 1.49%). Stage 2 extends this to cross-month and within-site scene variation: Oct 12 Aisle_CW (cross-month, four months later, $\Delta\text{APE} = 0.000$ mm, 2/64 masked, max 4.89%) and Oct 12 Hallway_Full_CW_Run_2 (first Hallway dynamic mask test, $\Delta\text{APE} = 0.000$ mm, 3/64 masked, max 0.63%). This upgrades to an 11/16-config, 5-session evidence chain (same-day + cross-day + cross-month + Hallway within-site variation). A follow-up high-coverage selective-mask pressure test on Jun 15 Aisle_CCW_Run_1 masks 5/64 frames with max per-frame coverage 17.32% and mean coverage 0.568%; it remains small in absolute terms but no longer exactly neutral ($\Delta\text{APE} = +0.114$ mm, $\Delta\text{RPE} = -0.006$ mm). Thus, selectivity is necessary but not sufficient by itself: coverage magnitude also defines the boundary between trajectory-neutral and mildly perturbed behavior.

TABLE VII
PARAMETER ABLATION SUMMARY (P154; DEFAULT: MIN_SESSIONS=2, MIN_FRAMES=4, MAX_DYNAMIC_RATIO=0.20, MIN_SUPPORT=6, MIN_LABEL_PURITY=0.70).

Criterion	Sweep	Retained	Sensitivity
min_sessions	1 / 2 / 3	7 / 5 / 5	High at 1→2
min_frames	2 / 4 / 6	8 / 5 / 5	High at 2→4
max_dynamic_ratio	0.10 / 0.20 / 0.30	5 / 5 / 5	Saturated

TABLE VIII
BASELINE COMPARISON: NAIVE (B0), PURITY/SUPPORT HEURISTIC (B1), FULL ADMISSION CONTROL (B2).

Metric	B0 (Naive)	B1 (Purity/Support)	B2 (Full Control)
Total admitted	20/20	13/20	5/20
Forklift admitted	4/4	4/4	0/4
Phantom risk	20.0% (4/20)	21.1% (4/19)	0.0% (0/5)
Infrastructure retention	5/9 (55.6%)	5/9 (55.6%)	5/9 (55.6%)
Reduction over B0	—	−35.0%	−75.0%
McNemar vs B2	$p<0.0001$	$p=0.125$ (n=4)	—

G. Admission Criteria Defense: Ablation, Baseline Comparison, and Per-Category Analysis

This subsection tests whether the five-criterion admission policy is doing substantive work beyond simple detection confidence or support counts. The ablation isolates which thresholds matter, the baseline comparison shows why purity/support heuristics admit forklifts, and the per-category analysis checks that the final map composition remains interpretable rather than merely smaller.

VIII. DISCUSSION

A. Why Boolean Criteria (Not Learned)

The five admission criteria (§V-E) are deliberately boolean rather than learned. Three considerations motivate this choice.

Traceability. Every admission or rejection decision maps to a specific criterion violation, enabling per-object provenance auditable without model introspection. Learned classifiers (e.g., a neural admission gate trained on object features) would obscure the decision boundary.

Within-site variation. Boolean thresholds on session count, label purity, and dynamic ratio remain stable across the Aisle and Hallway protocols without retraining. This supports within-warehouse floor-plan variation, not cross-site generalization: a learned gate trained only on forklift-rich Aisle data might specialize to that scene and fail on the Hallway negative-control environment.

Domain-informed priorities. The weights in the trust score (1) reflect domain priorities—session count (α=0.4) is weighted more heavily than observation support (β=0.3) because cross-session persistence is a stronger indicator of infrastructure than repeated detection within a single session—rather than being optimized on a validation set. The dynamic-ratio weight (γ=0.5) intentionally penalizes dynamic contamination to produce a clean floor: $\tau(O) \leq 0.15$ for forklifts, $\tau(O) \geq 0.4$ for infrastructure.

TABLE IX
PER-CATEGORY RETENTION AND REJECTION MATRIX (AISLE LADDER, N=20).

Category	Clusters	Retained	Rejected	Retention Rate
barrier	5	2	3	40.0%
work table	5	2	3	40.0%
warehouse rack	3	1	2	33.3%
forklift	4	0	4	0.0%
unclassified	3	0	3	0.0%

B. Dynamic-Ratio Bimodality

The observed separation—infrastructure clusters at dynamic ratio = 0.00, forklifts at $\geq$0.83, with no clusters in the intermediate range (0.01–0.82)—is a striking feature of the data. This bimodality justifies the insensitivity of `max_dynamic_ratio` across a wide range (0.10–0.30) shown in the parameter ablation (§VII-G).

However, this bimodality may partially reflect the frontend state classifier's decision boundary rather than purely physical separation. The detector reports three states (`static`, `candidate`, `dynamic_agent`); if the classifier assigns `dynamic_agent` conservatively, dynamic objects receive consistently high ratios while static objects receive zero. This caveat is noted as a limitation (§IX, item 4): the bimodality is real in the data but its sharpness may be frontend-dependent. Multi-frontend ablation would be needed to quantify this dependence.

C. Relationship to Landmark Selection in Classical SLAM

The admission-control problem has conceptual precursors in classical SLAM. EKF-SLAM feature management, PTAM map-point culling, and ORB-SLAM's map-point policies all implement observation-quality thresholds for geometric landmarks. ORB-SLAM3's map-merging capability further introduces cross-session geometric verification. Our admission criteria carry this principle into the semantic domain: just as ORB-SLAM culls map points with low observation count or high reprojection error, our framework culls semantic objects with low session support or high dynamic ratio. The key difference is that semantic admission requires reasoning about object identity persistence rather than geometric reprojection error, and the criteria must distinguish between static infrastructure and structurally similar dynamic agents (forklifts that look geometrically stable but are semantically inadmissible). RTAB-Map's memory management similarly filters based on retrieval frequency; our session-count criterion is a semantic analogue.

D. Single-Site Limitation and Cross-Site Validation

The current evidence is limited to a single warehouse site (TorWIC). While the Aisle and Hallway protocols provide within-site protocol-level diversity, external validation at other industrial sites (different warehouse layouts, lighting conditions, object categories) is deferred. A multi-site validation would require: (1) a multi-site industrial SLAM dataset with ground-truth object annotations across revisits (no such public dataset exists to our knowledge); (2) recalibration of the detection prompt set for site-specific object categories; and (3) verification that the admission thresholds (`min_sessions`=2, `max_dynamic_ratio`=0.2) transfer without retuning. Until such data is available, the current evidence is bounded to the TorWIC environment, as noted among the limitations (§IX, items 1–2).

IX. LIMITATIONS

1) **Evaluation scale.** The admission-control evaluation operates on 20 candidate clusters (across all protocols after deduplication). Bootstrap 95% confidence intervals for per-protocol admission rates range from 18.2–72.7% (same-day, n=11) to 31.2–81.2% (Hallway, n=16); the pooled rate CI is 37.3–64.7% (n=51). Hallway versus Aisle admission rates do not differ (Fisher exact $p = 0.76$). Between-baseline McNemar exact tests (Aisle ladder, paired n=20): B0/B1 $p<0.001$, B1/B2 $p=0.125$ (n=4 discordant pairs), B0/B2 $p<0.0001$. B1/B2 is not significant at $\alpha=0.05$ but the effect direction is deterministic (B2 never admits a cluster B1 rejects; all 4 discordant pairs are B1-admitted/B2-rejected forklifts). Statistical power is limited by cluster count; confidence intervals and p-values should be interpreted with appropriate caution.
2) **Single industrial site.** TorWIC is collected in a single warehouse. Although Aisle and Hallway provide protocol-level diversity, external validation at other industrial sites is deferred.
3) **Detection confidence not logged.** Per-detection NN confidence scores were not preserved during observation extraction. The label-purity proxy is used as a documented substitute.
4) **Dynamic ratio may depend on frontend.** The dynamic ratio is computed from the detector's state classifier (static / candidate / dynamic_agent). The observed bimodality (0.00 vs $\geq$0.83) may partially reflect the classifier's decision boundary rather than purely physical separation. The insensitivity of max_dynamic_ratio to threshold choice is demonstrated through ablation but the bimodality's dependence on frontend behavior is a caveat.
5) **No comparison against published map-management policies.** The baseline comparison (naive-all-admit, purity/support heuristic) uses deliberately simple policies to isolate the contribution of individual signals. Comparison against map-management policies in published SLAM systems (e.g., RTAB-Map's memory management, ORB-SLAM3's map-point culling) is deferred to future work.
6) **Single-session infrastructure loss.** Infrastructure appearing in only one session cannot be verified for cross-session persistence and is conservatively rejected. This is a design choice—precision over recall for map entries—but entails genuine infrastructure omission.
7) **Dynamic SLAM boundary.** evo APE evaluation across the bounded DROID-SLAM chain reveals three regimes.

Selective low-coverage semantic masks are trajectory-neutral ($|\Delta\text{APE}| \leq 0.006$ mm in the original 7/12 neutral configs, extended by four neutral replication sessions). Aggressive mask strategies (temporal/flow propagation, uniform first-N masks) introduce measurable perturbations (0.92–7.52 mm, 5 configs). A high-coverage but still selective Jun 15 Aisle_CCW pressure test (max 17.32% frame coverage, mean 0.568%) produces only a small perturbation ($\Delta\text{APE} = +0.114$ mm). Mask selectivity is therefore necessary, but high dynamic coverage can still move the trajectory at sub-millimetre scale. Bootstrap 95% CI for the neutral rate is 43.8–87.5% (11/16 configs); the two-group ΔAPE separation at $|\Delta\text{APE}| \leq 0.006$ mm is complete (gap=0.914 mm, no overlap). We do not claim navigation benefit or full-map quality improvement from masking.

8) **Backend boundary and ORB-SLAM3 caveat.** The trajectory-perturbation evidence remains DROID-SLAM-primary. A follow-up ORB-SLAM3 mono cross-check on Jun 15 Aisle_CCW produced only sparse keyframe trajectories (raw 5 keyframes, masked 9 keyframes over roughly 1.2 s of a 64-frame, approximately 7 s window). evo APE/RPE are mathematically valid on those matched keyframes (raw/masked APE RMSE 0.041/0.043 m), but the sample is underpowered and not directly comparable to DROID-SLAM dense per-frame trajectories. We therefore treat ORB-SLAM3 only as a low-power backend-risk check and make no raw-vs-masked improvement, trajectory-neutrality generalization, or cross-backend performance claim.
9) **Ground-truth object annotations.** TorWIC provides ground-truth object IDs; we use them for protocol configuration but the framework itself operates without ground truth.

X. Conclusion

This paper presents a systematic, evidence-backed approach to session-level map admission control for semantic-segmentation-assisted SLAM in dynamic industrial environments. The framework transforms open-vocabulary RGB-D segmentation outputs into observation, tracklet, map-object, and revision layers, then applies explicit persistence, stability, and dynamicity signals to retain stable semantic landmarks and suppress dynamic contamination.

Current TorWIC evidence documents a reproducible evidence chain across same-day, cross-day, and cross-month Aisle protocols, with Hallway retained as a separate within-site variation branch. Forklift-like evidence is consistently rejected as dynamic contamination (50%–71% rejection share), while barriers, work tables, and warehouse racks are selectively retained based on multi-session evidence, label consistency, and static dominance.

The dynamic SLAM backend evaluation identifies a boundary condition: selective low-coverage semantic masks are trajectory-neutral for DROID-SLAM across the original single-session sweep and four replication sessions, while aggressive mask strategies (propagation, uniform first-N) introduce measurable perturbations (0.92–7.52 mm). A high-coverage selective-mask pressure test remains far below the aggressive-mask failures but produces a small nonzero shift ($\Delta\text{APE} = +0.114$ mm), showing that both selectivity and coverage magnitude matter for bundle adjustment. The admission criteria are validated through parameter ablation across 27 parameter combinations, baseline comparison (demonstrating that purity/support heuristics fail to reject any forklift), and per-category retention/rejection analysis.

The contribution is bounded: not a lifelong SLAM benchmark, not dense dynamic reconstruction, and not a downstream navigation-gain claim. It is a systematic map maintenance methodology — five boolean criteria with per-object provenance at every layer — supported by reproducible evidence and a complete supplementary evidence package that enables per-cluster auditability of all admission and rejection decisions.

Industrial Long-Term Semantic Mapping: Finalizing Figure v3

First-8-frame summary

Session	Frames	ObsTotal	Tracklets	MapObjects	DynamicAgents	Labels
Jun15 Aisle_CV first8	8	38	38	43	9	fork, forklift, work, work table, yellow barrier
Jun23 Aisle_CV first8	8	81	81	89	19	fork, forklift, warehouse rack, work table, yellow barrier
Oct12 Aisle_CV first8	8	44	44	50	20	barrier, fork, forklift, table, work table, yellow barrier

Cross-session first-8 clusters

Cross-session Cluster Summary (First 8 Frames, Real Industrial Runs)

inputs=24 frame_level_objects=182 clusters=14

Cluster	Label	State	Sessions	Frames	Support	Raw labels
cluster_0004	work table	candidate	3	24	44	work table:44
cluster_0003	forklift	dynamic_agent	2	16	24	forklift:23, fork:1
cluster_0001	yellow barrier	candidate	2	16	18	yellow barrier:18
cluster_0002	yellow barrier	candidate	2	16	18	yellow barrier:18
cluster_0006	warehouse rack	candidate	1	8	18	warehouse rack:18
cluster_0012	forklift	dynamic_agent	1	8	12	forklift:11, fork:1
cluster_0010	forklift	dynamic_agent	1	8	9	forklift:9
cluster_0007	warehouse rack	candidate	1	8	9	warehouse rack:9
cluster_0014	yellow barrier	candidate	1	8	9	yellow barrier:7, barrier:2
cluster_0011	yellow barrier	candidate	1	7	8	yellow barrier:8
cluster_0008	forklift	dynamic_agent	1	6	6	forklift:5, fork:1
cluster_0009	work table	candidate	1	3	3	work table:3

Stable selection v5

Stable Object Selection v5 (cross-session first-8-frame, label-purity aware)

criteria: sessions>=2 frames>=4 support>=6 dyn<=0.2 purity>=0.7

Selected stable objects

label	support	sessions	frames	dyn_ratio	purity	raw labels
work table	44	3	24	0.00	1.00	work table:44
yellow barrier	18	2	16	0.00	1.00	yellow barrier:18
yellow barrier	18	2	16	0.00	1.00	yellow barrier:18

Rejected clusters

label	support	sessions	frames	dyn_ratio	purity	raw labels
forklift	24	2	16	0.36	0.96	forklift:23, fork:1
warehouse rack	18	1	8	0.00	1.00	warehouse rack:18
forklift	12	1	8	0.92	0.92	forklift:11, fork:1
forklift	9	1	8	1.00	1.00	forklift:9
warehouse rack	9	1	8	0.00	1.00	warehouse rack:9
yellow barrier	9	1	8	0.00	0.78	yellow barrier:7, barrier:2
yellow barrier	8	1	7	0.00	1.00	yellow barrier:8
forklift	6	1	6	0.83	0.83	forklift:5, fork:1
work table	3	1	3	0.00	1.00	work table:3
work table	3	1	2	0.00	1.00	table:3

Stable subset

Stable Object Subset (Current Long-Term Map Core)

num_stable_objects: 3

ID	Label	Support	Sessions	Frames	Raw labels	Center
stable_001	yellow barrier	18	2	16	yellow barrier:18	[1177.2, 546.9, 0.0]
stable_002	yellow barrier	18	2	16	yellow barrier:18	[300.3, 462.8, 0.0]
stable_003	work table	44	3	24	work table:44	[281.1, 349.9, 0.0]

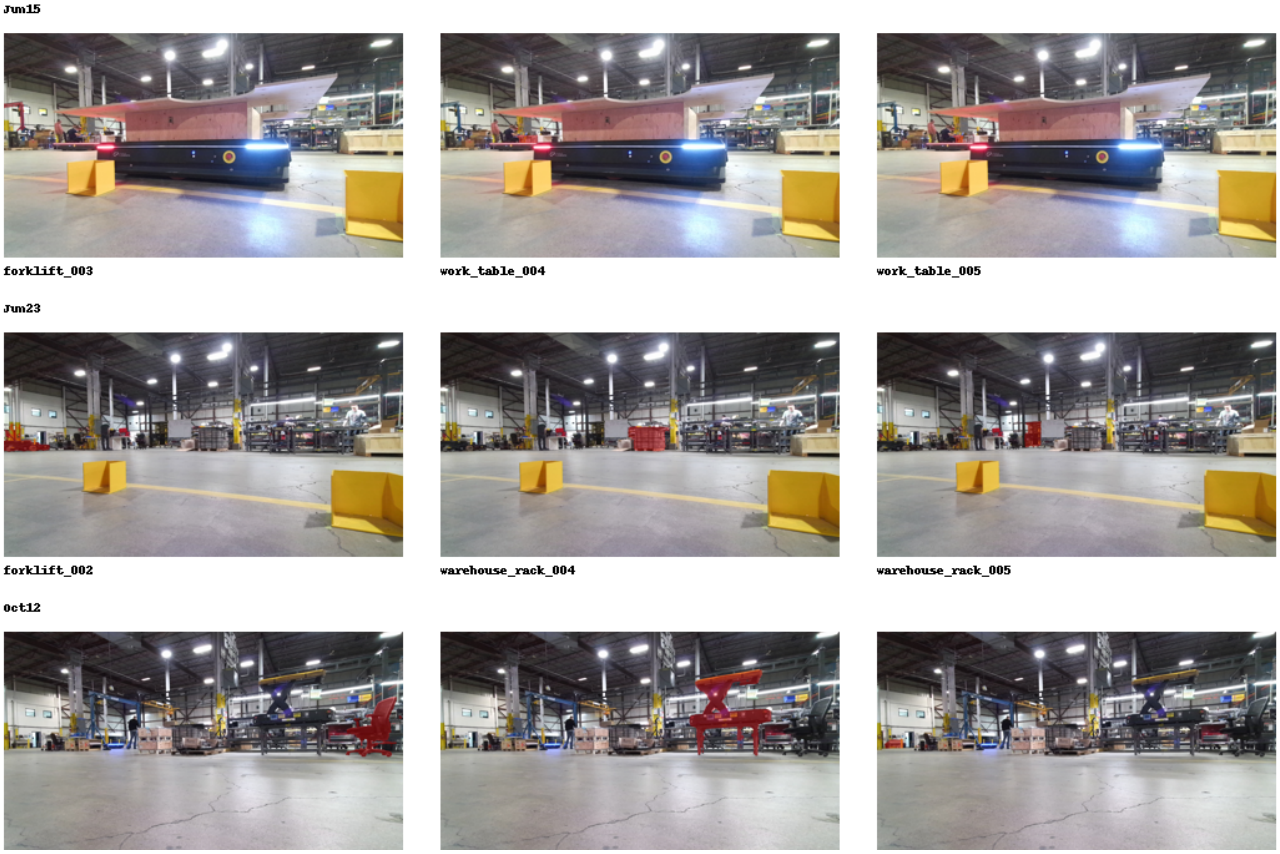

Fig. 2. Four-protocol rejection-ratio overview: same-day, cross-day, cross-month Aisle protocols and Hallway within-site scene variation. Retained infrastructure objects (barriers, work tables, warehouse racks) are shown in green; rejected clusters (forklifts, single-session, low-quality) in red.

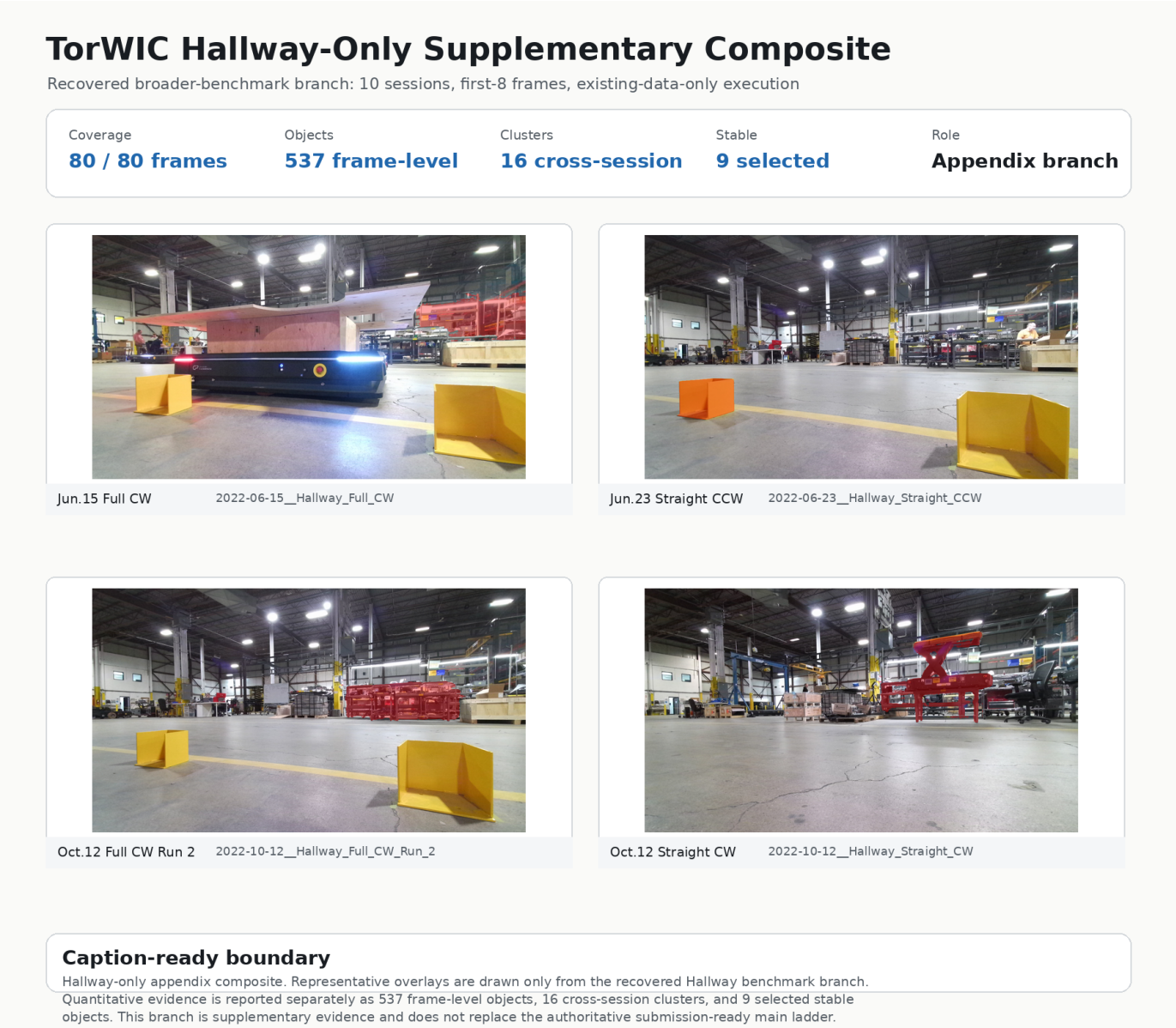

Fig. 3. Hallway within-site scene variation: 10 sessions across 3 dates, 537 observations, 16 clusters, 9 retained. The Hallway is a different floor plan within the same TorWIC warehouse, serving as a within-site negative control—no forklifts operate in this scene, and the admission framework correctly rejects all dynamic candidates.

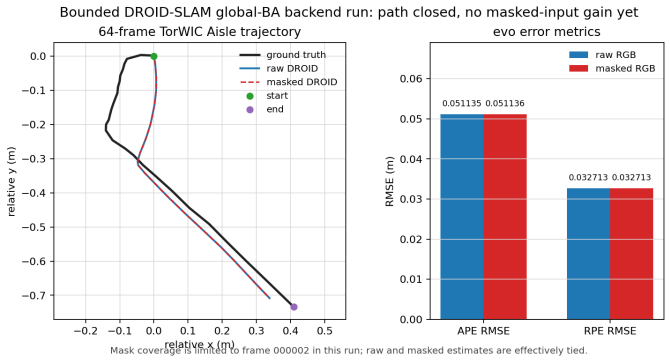

Fig. 4. Trajectory comparison: raw RGB (baseline) vs semantic-masked RGB (P135). Both trajectories are visually and metrically indistinguishable (ΔAPE $= 0.000$ mm).

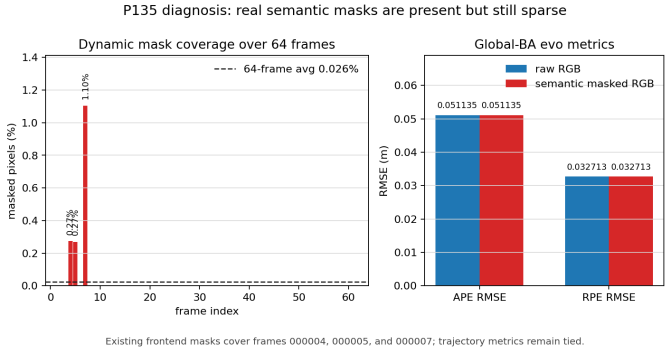

Fig. 5. Per-frame dynamic mask coverage distribution (64-frame window, Aisle_CW_Run_1). Selective semantic frontend masks cover only frames containing verified dynamic agents.

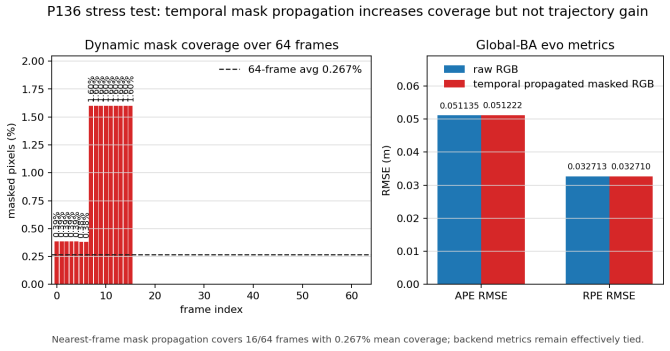

Fig. 6. Temporal propagation stress test (P136). Masks propagated temporally beyond actual dynamic frames; ΔAPE $= +7.517$ mm.

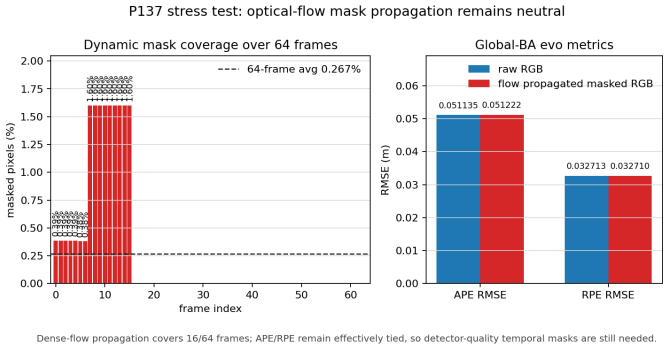

Fig. 7. Optical-flow propagation stress test (P137). Masks propagated via flow; ΔAPE $= +7.517$ mm, identical to temporal propagation.

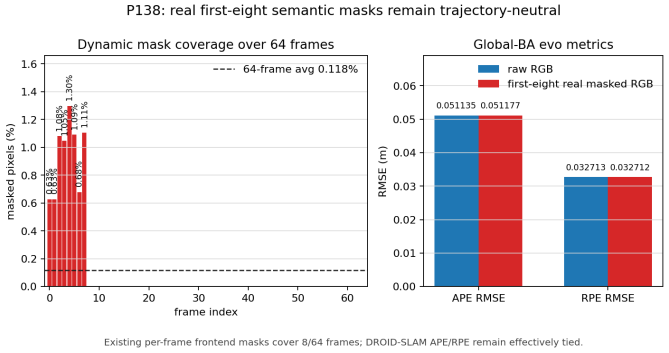

Fig. 8. First-8 uniform mask diagnostic (P138). ΔAPE $= +0.922$ mm.

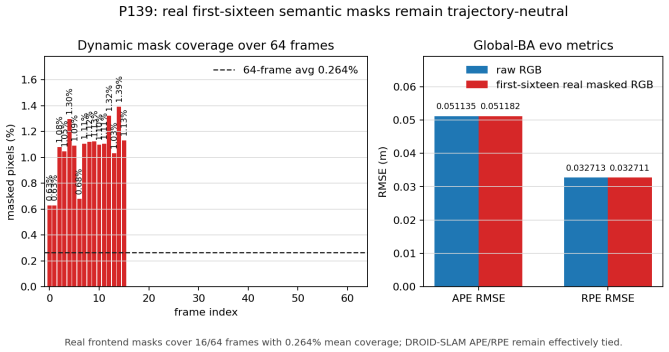

Fig. 9. First-16 uniform mask diagnostic (P139). ΔAPE $= +0.921$ mm—plateau confirms initialization sensitivity.

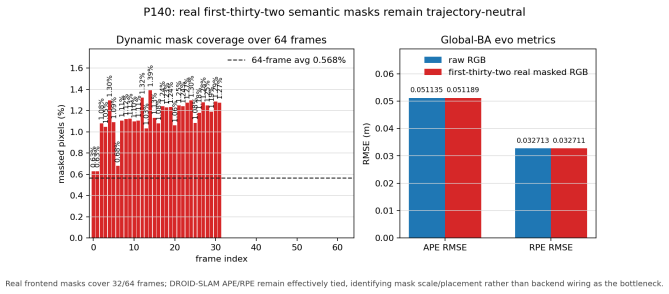

Fig. 10. First-32 uniform mask diagnostic (P140). ΔAPE $= +0.920$ mm—further plateau.

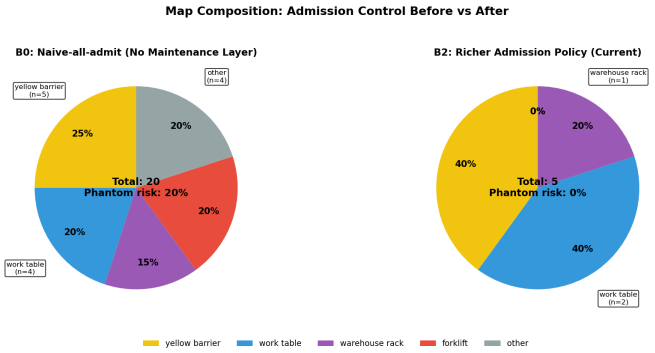

Fig. 11. B0 (naive) vs B2 (full admission control) map composition. B0 admits all 20 clusters including 4 forklifts (phantom risk 20.0%); B2 retains 5 infrastructure-only clusters (phantom risk 0.0%).

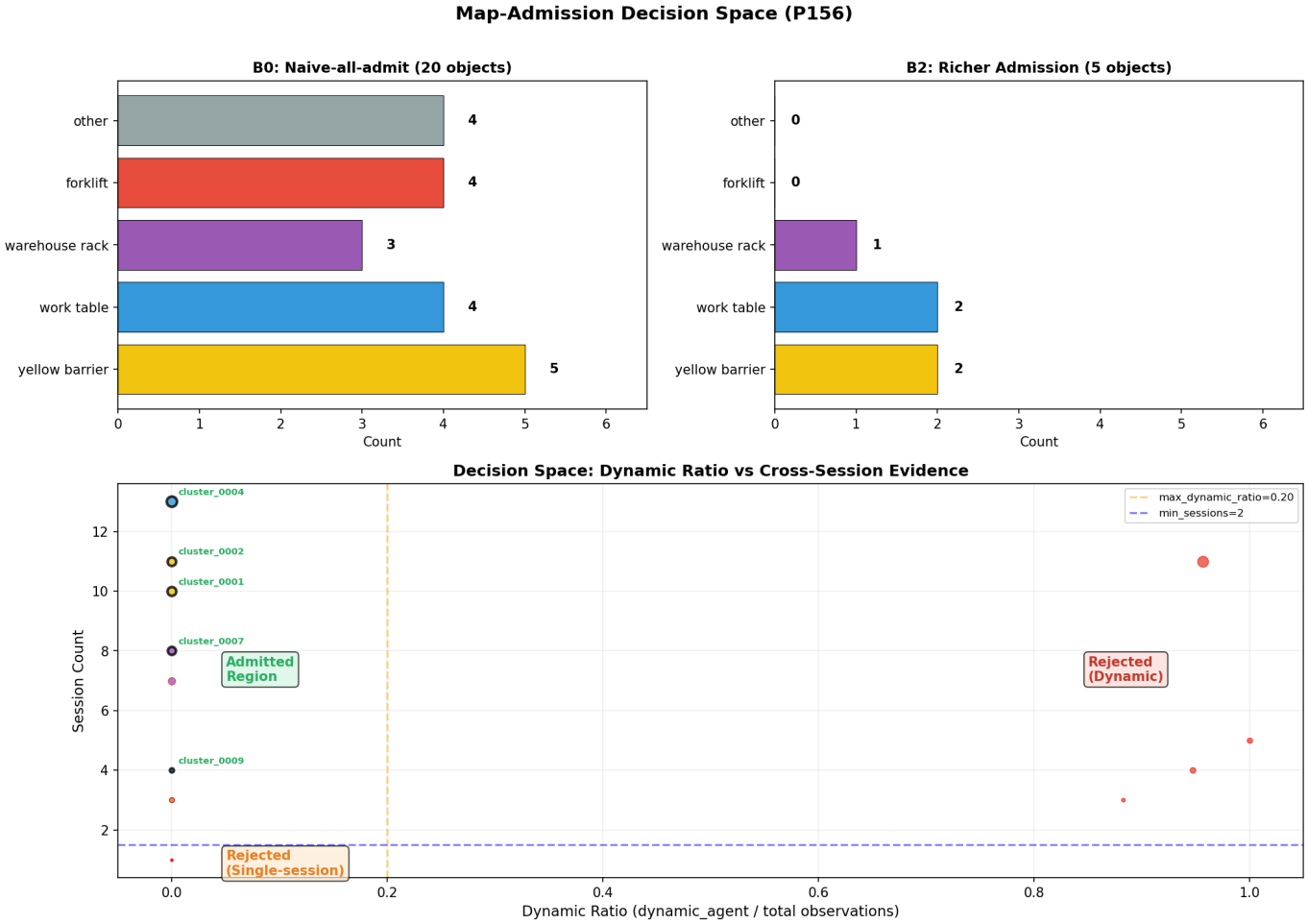

Fig. 12. Admission decision space: dynamic_ratio $\times$ session_count scatter for all 20 Aisle clusters. Infrastructure clusters (barriers, work tables, racks) cluster at dynamic ratio $= 0.00$, session count ≥ 2; forklifts cluster at dynamic ratio ≥ 0.83. The clear separation justifies the max_dynamic_ratio threshold insensitivity shown in parameter ablation.

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